As mentioned, the primary significant defect in TOF is an obstruction, specifically **pulmonary valve stenosis**, which is a form of RV outflow tract (RVOT) obstruction. The degree of this obstruction will contribute to how and when a child develops symptoms of TOF. In severely restricted pulmonary artery flow, young neonates will develop cyanosis and require surgical intervention very early in life.

CLINICAL CORRELATION

Pulmonary valve stenosis is a CHD that can occur as part of TOF, but it more commonly exists in isolation. The symptoms depend on how severe the obstruction is, but commonly include an **ejection systolic, crescendo-decrescendo murmur** at the left upper sternal border**.** Severe isolated stenosis can cause right heart failure with dyspnea on exertion.

Children with less significant obstruction may present with **tet spells** (episodes of cyanosis, rapid breathing, and irritability). These typically happen when the RVOT obstruction suddenly worsens, which can occur for various reasons including agitation, feeding, or illness (fever, anemia, hypovolemia). Tet spells are unique to TOF, and a child having one should receive emergent intervention and further investigation.

Pathophysiology

Mnemonic

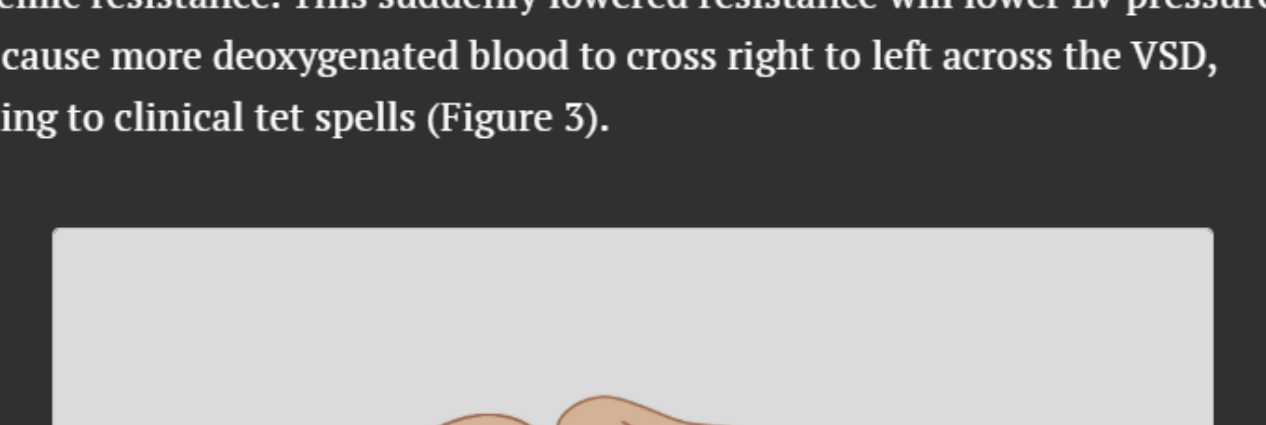
To remember the four components of tetralogy of Fallot, think **PROV**: **P**ulmonic stenosis, **R**ight ventricular hypertrophy, **O**verriding aorta, and **V**entricular septal defect.

Embryology. TOF is thought to be due to the malalignment of the conal septum. This malalignment causes the outflow tract to stenose, the aorta to override, and VSD to persist. The RV hypertrophy persists after birth. In the fetus:

- The pulmonic stenosis in TOF is due to thickening of either the pulmonary valve or the proximal pulmonary artery wall.
- The RV hypertrophy is an adaptive response to the RVOT obstruction, since the RV muscle has to work harder to push blood through the pulmonic stenosis.
- The overriding aorta arises from abnormal development of the spiral septum (which divides the truncus arteriosus into the aorta and pulmonary artery) and involves abnormal migration of neural crest cells. The aorta thus arises from both the RV and LV.
- The VSD arises from the incomplete development of both the membranous and muscular sep[illegible] migration of neural crest cells.

Why does deoxygenated blood from the RV not mix with oxygenated blood in the LV in a newborn with an isolated VSD?

Complications. What accounts for tet spells? One common explanation is that during exertion (e.g., feeding or crying), the systemic arteries dilate, lowering systemic resistance. This suddenly lowered resistance will lower LV pressure and cause more deoxygenated blood to cross right to left across the VSD, leading to clinical tet spells (Figure 3).

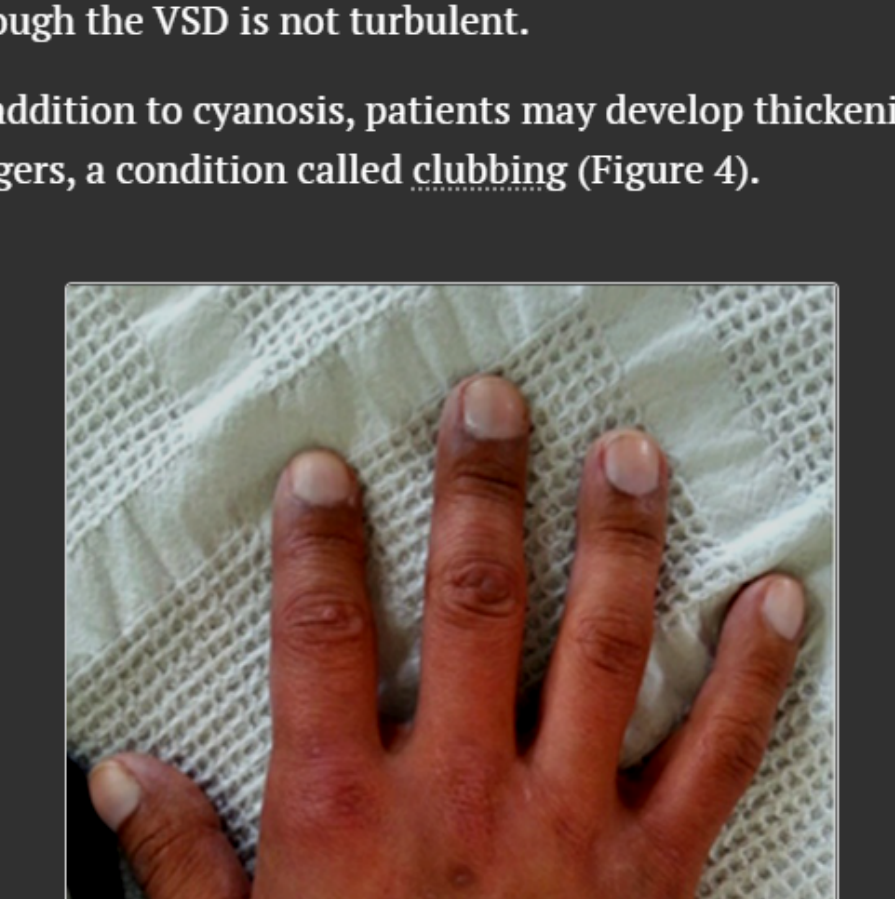


RVOT obstruction.

On auscultation, the patient may only have a **single S_2** from the normal aortic valve, as the pulmonic component is usually not heard because of pulmonary stenosis. This would most likely be detected by auscultating the heart during inspiration, when a normal physiologic splitting of S_2 would be expected but, in this case, would not occur.

A systolic ejection murmur can be heard at the upper left sternal border. The turbulent blood flow through the RVOT obstruction causes this, not the VSD, since flow through the VSD is not turbulent.

Over time, in addition to cyanosis, patients may develop thickening of the end part of the fingers, a condition called clubbing (Figure 4).



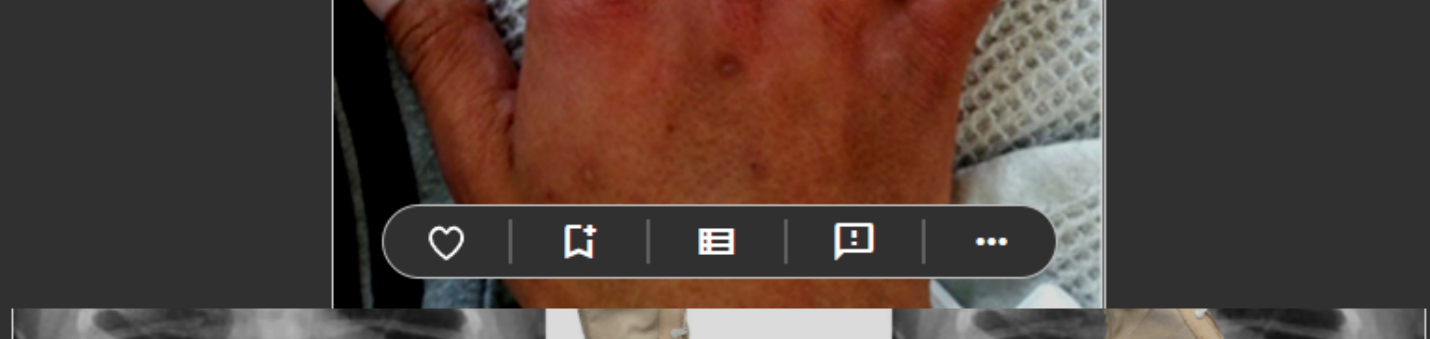
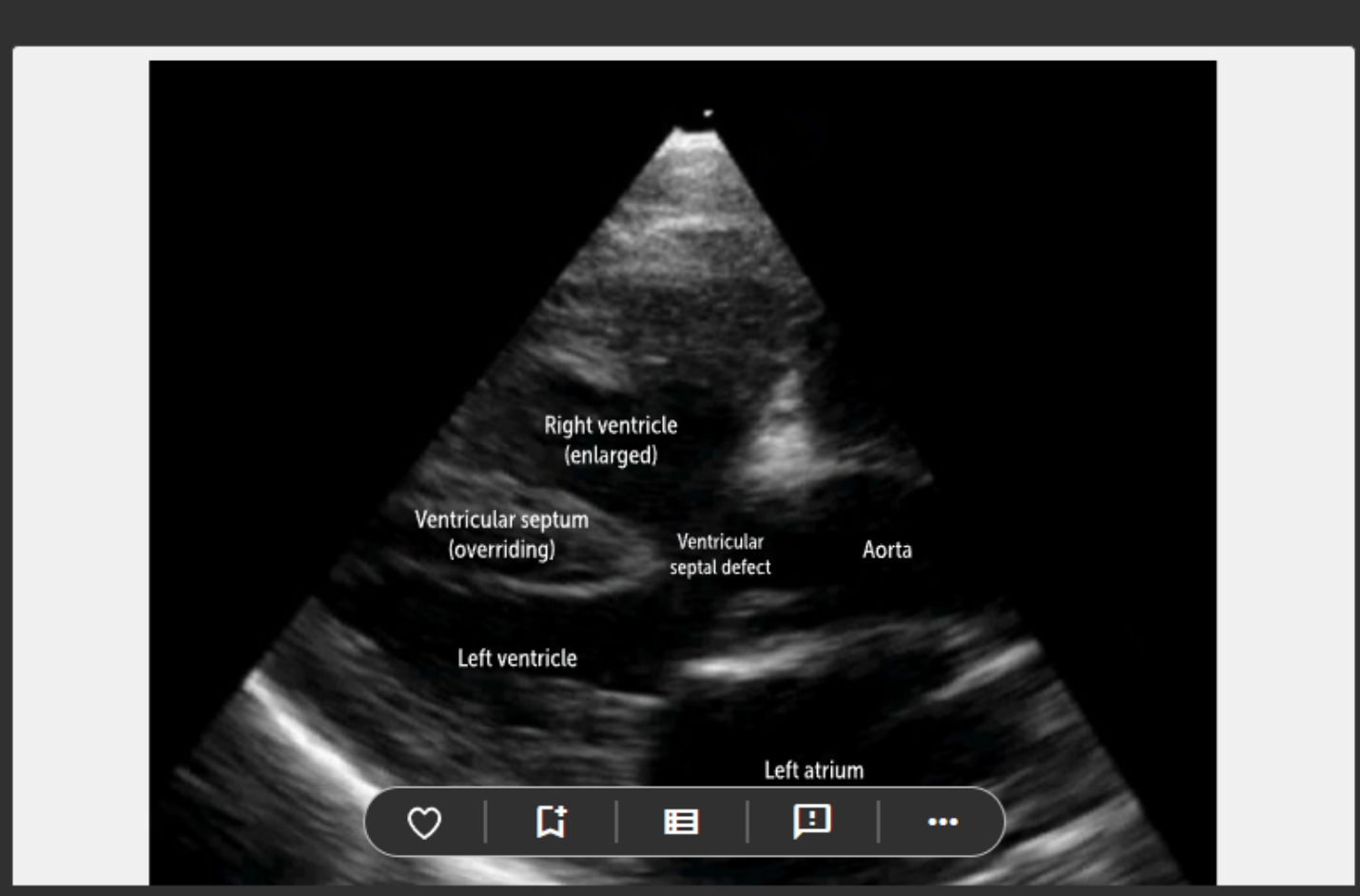


Figure 5 Chest x-ray in tetralogy of Fallot showing boot-shaped heart

Echocardiography. Echocardiography is the definitive diagnostic test for the assessment of TOF. It shows the VSD and can also be used to evaluate the RVOT obstruction (Figure 6).


Right ventricle (enlarged)
Ventricular septum (overriding)
Ventricular septal defect
Aorta
Left ventricle
Left atrium

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Acyanotic Congenital Heart Defects (Left–to–Right Shunts)

Last updated February 28, 2026 · 21 min read
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Adapted by: Alexis Gillett, Rhome Hughes, Stephen Nix, Chelsey Deel
Editors: Russell Johnson, Jordan Finn
Contributor: Nancy Halliday
Reviewers: Diptiman Bose, Carrie Elzie

Adapted from: Patent Ductus Arteriosus

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Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of acyanotic congenital heart defects.

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are isolated lesions.

Common risk factors include maternal rubella infection during pregnancy, birth at high altitude, and the presence of other CHDs. PDA is also commonly seen in premature infants, especially those with respiratory distress syndrome and very low birth weights.

What are the major risk factors associated with PDA?

Clinical Presentation

The severity of this condition depends on the diameter of the PDA and the amount of blood shunting through it: larger diameters allow more blood to be shunted and are associated with more significant symptoms. Less severe PDAs are often asymptomatic at birth and present later in childhood with symptoms such as fatigue and e[illegible]

imaging techniques like echocardiography with Doppler to assess the size and impact of the shunts.

- Management strategies vary based on defect size and symptoms, ranging from monitoring and medical management to surgical or catheter-based interventions for larger or symptomatic defects.

Review Questions

1. A 7-year-old boy presents to his pediatrician with exertional dyspnea, fatigue, and cyanosis of the lower extremities. Which of the following is NOT a risk factor associated with this congenital heart disease?

A. ○ Birth at high altitude
B. ○ Congenital rubella infection
C. ● Maternal lithium use
D. ○ Premature birth
E. ○ Presence of other congenital heart defects

Hide Explanation

The correct answer is maternal lithium use (C). The patient is presenting with classic symptoms of a patent [illegible] use is associated with apical displacem[illegible]y), but not PDA. Birth [illegible] diagnose PDA, but echocardiography is preferred because it is less costly and provides additional functional information that can be used to rate the severity. Further, avoiding ionizing radiation exposure would be especially desirable in a patient of this age. Electrocardiogram (D) and magnetic resonance imaging (E) can also detect findings associated with PDA, but neither is the preferred modality.

3. Which acyanotic CHD is characterized by a left-to-right shunt through an interventricular opening?

A. ○ Atrial Septal Defect
B. ● Ventricular Septal Defect
C. ○ Patent Ductus Arteriosus
D. ○ Tetralogy of Fallot

Hide Explanation

The correct answer is Ventricular Septal Defect (B). VSD involves a left-to-right shunt through an interventricular opening. Atrial Septal Defect (A) involves an interatrial opening. Patent Ductus Arteriosus (C) is a shunt via the ductus arteriosus. Tetralogy of Fallot (D) is a cyanotic defect, not acyanotic.

4. What is a major complication of untreated large Atrial Septal Defect (ASD)?

A. ● Pulmonary hypertension

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of acyanotic congenital heart defects.~~

Objective Confidence: 1/1

complications.

Prognosis. Patients with TOF generally do not do very well without surgery. Approximately 50% of patients without treatment do not survive past age 3 years.

What Is Tricuspid Atresia? (LO#1)

As its name suggests, tricuspid atresia refers to the absence of a functional tricuspid valve. This means no direct connection between the right atrium (RA) and RV. In fetal circulation, the foramen ovale allows blood to flow from the right atrium to the left atrium, but unless the foramen ovale remains patent or another atrial septal defect develops, this anomaly is incompatible with life. Additionally, unless another cardiac defect permits left-to-right shunting past the nonfunctional tricuspid valve (like a ventricular septal defect or patent ductus arteriosus), blood will have no way to reach the lungs and will quickly be fatal after birth.

Like TOF, this is a right-sided obstruction and is a **cyanotic** CHD with a **right-to-left shunt**. Therefore, poorly oxygenated blood circulates in the systemic circulation.

TA is less common than TOF, with an estimated incidence of 1 in 10,000 live births.

Clinical Presentation

As stated, blood must be able to shunt from the right atrium to the left atrium to be compatible with [illegible]ght shunt must be present that allows blood from the left ventricle to supply the pulmonary

2022.
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